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(1) Research Program/Project/Study Title: Extraction And Characterization Of Plant- Based Nonstarch Polysaccharides (NSPs) For Beverage Application	
Proponent Name:	Marife R. Hilapad
Academic Rank:	Associate Professor V
Co-Authors (with academic rank):	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	Marife R. Hilapad
Academic Rank:	Associate Professor V

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	Mambusao
		Province:	Capiz
		Region:	6
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity FOOD/ Okra, Lemon grass, Red joy sessile and Calamansi			(8) Discipline/Field of Specialization FOOD SCIENCE AND TECHNOLOGY

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Non starch polysaccharides (NSPs) are carbohydrate fractions excluding starch and free sugars. These are polymeric carbohydrates differing in composition and structure from amylase and amylopectin (Chesson, 2001; Sethy et al, 2015). The major polysaccharides of NSPs are cellulose, pectins, β -glucans, pentosans, heteroxylans, and xyloglucan (Englyst, 1989, Vikas et al, 2012). Mucilages, alginates, exudates gums, and various modified polysaccharides are also constituents of the nonstarch polysaccharides (Asp et al., 1983). NSPs through their high water binding capacity play an important role in providing bulk to gut contents and allow easy passage through the human intestine (Davidson and McDonald, 1998), thereby playing a crucial role for the correct functioning of the digestive system. A major proportion of dietary NSPs escapes the small intestine nearly intact, and is fermented into short chain fatty acids (SCFAs) by commensal microflora residing in the colon and cecum and promotes normal laxation. Short chain fatty acids have a number of health promoting effects such as lowering the pH of the colon, inhibiting growth of pathogenic organism (Scott et al., 2008), increasing mineral absorption, maintaining normal bowel structure and function, preventing or alleviating colonic-based diarrhea, and stimulating colonic blood flow and fluid and electrolyte uptake.

Okra (*Abelmoschus esculentus*) pod is an important vegetable in the tropics and subtropics (Jideani & Bello, 2009). It contains mucilage which is thick slimy polysaccharides and is used to thicken soups and stews (Ahiakpa et al., 2014; Biswal, Karna, & Patel, 2014), as whipping agent for reconstituted egg whites, an additive in the formulation of flour-based adhesives and as additive for clarifying sugarcane juice (Gemedede et al, 2018). Likewise, mucilage from okra is medically proven to be linked to anticancer, antimicrobial, hypoglycemic, anti-ulcer activities (Ansari, Houlihan, Hussain, & Pieroni, 2005), as well as its ability to bind cholesterol and bile acid carrying toxins by filtering the liver (Shui & Peng, 2004). Moreover, pods of okra contain a desirable amount of mucilage contents and are potential sources of natural antioxidants (Gemedede et al, 2018).

On the other hand, a non-alcoholic drink product known as a functional beverage is made with ingredients such as raw fruits, herbs, vitamins, minerals, amino acids, and other bioactive compounds that have been shown to have specific health advantages. Functional beverages have grown in popularity over the past few years because of their numerous particular health advantages (Shukla and Salve, 2023). The combination of lemon grass and red joy sessile leaves in calamansi juice drink is a healthy beverage developed at CAPSU- MSC. It is carefully processed to make sure that the bioactive components of the raw materials are preserve. However, through time, sediments are formed at the bottom of the bottle making it appear less attractive to the consumers.

Thus, the addition of okra mucilage in blended beverage formulation will not only improve the clarity of the juice but will also provide additional nutritional, medicinal and health benefits to the product.

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(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned. .)

This study will make use of Nonstarch Polysaccharides, particularly the okra mucilage in improving the quality of the CAPSU-MSC developed beverage. The mucilage will be extracted from okra, it will then be characterized to determine its functional properties, and added to blended beverage. The maturity of okra in relation to its mucilage content will be investigated and its stability when incorporated in the juice will be analyzed. The finished product will be subjected to physical, chemical and microbial analyses to ensure its quality and safety. Proximate composition, antioxidant activity and shelf life studies will be conducted on the optimized product only. Assessing the consumer acceptability and cost analysis will form part of the study.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

The study generally aimed to extract and characterize the nonstarch polysaccharide from okra for beverage application.

The specific objectives are:

1. To determine the NSPs content of okra pod as affected by levels of maturity;
2. To evaluate the functional properties of NSPs from okra;
3. To optimize the levels of NSPs extracted from okra for blended beverage application;
4. To determine the sensory properties and acceptability of NSP-enriched blended beverage;
5. To evaluate the physical, chemical and microbiological characteristics of the formulated beverage.
6. To determine the antioxidant activity of the optimized blended beverage;
7. To conduct shel-life studies on the optimized product; and
8. To assess consumer acceptability of the optimized product.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

Nonstarch polysaccharides (NSPs) include all the plant polysaccharides other than starch. They are the key components of the cell walls of various grains and cover a great variety of biological functions and chemical structures. The major polysaccharides of NSP are cellulose, pectins, β -glucans, pentosans, heteroxylans, and xyloglucan. Non-starch polysachharides together form a major part of dietary fiber (DF) of grains and chemically DF is often defined as NSPs (Englyst, 1989, Vikas et al, 2012). According to Kumar et al (2012), Nonstarch polysaccharides (NSPs) occur naturally in many foods. The physiochemical and biological properties of these compounds correspond to dietary fiber. Nonstarch polysaccharides show various physiological effects in the small and large intestine and therefore have important health implications for humans. The remarkable properties of dietary NSPs are water dispersibility, viscosity effect, bulk, and fermentibility into short chain fatty acids (SCFAs).

Okra (*Abelmoschus esculentus*) pod is an important vegetable in the tropics and subtropics. Okra typically differs from most other common vegetables in having high mucilage content (Jideani & Bello, 2009). Okra mucilage has the potential for use as food, nonfood products, and medicine (Haruna, Aliyu, & Bala, 2016; Kumar et al., 2010). Okra mucilage is a mixture of natural polysaccharides, consisting of the monosaccharides D-galactose, L-rhamnose, and galacturonic acid associated with proteins and

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minerals(Anastasakis et al, 2009). According to Adetuyi and Dada (2014), the content of Zn and Ca in the mucilage of okra is higher than the mineral content of the whole okra fruit. Vegetable mucilages have been studied due to their biological activities in humans and animals. The polysaccharides from these mucilages demonstrate some important biological activities, such as immunomodulated and anti-inflammatory. The viscous nature of okra mucilage can greatly increase inhibition of both in vitro glucose trapping in the cell and the absorption of sugar from the intestine. This fact was confirmed by Chukwuma et al.(2018), who compared the anti-hyperglycemic activity of amadumbe (*Colocasia esculenta*) and okra mucilage and observed greater inhibition of glucose absorption by okra due to the higher viscosity of its mucilage. The natural polysaccharides found in plant mucilages have excellent antioxidant activity, which prevents cell damage caused by reactive oxygen species. In addition to acting on the sequestration of free radicals, these polysaccharides can increase the levels of superoxide dismutase (SOD), favoring the antioxidant mechanism. Okra mucilage, especially the pectic polysaccharide fraction WOP-2 (a rhamnogalacturonan I backbone with type II arabinogalactan side chains substituted partly at O-4 of rhamnopyranosyl), have antioxidant activity that can assist in decreasing lipid peroxidation reactions that cause the destruction of beta cells (Zhang et al, 2018)

Emulsifiers are one of the main classes of additives used in the food industry, and they play an important role in the formation and stability of emulsions, including milks, creams, seasonings, desserts, sauces, and beverages (Xu et al, 2019).

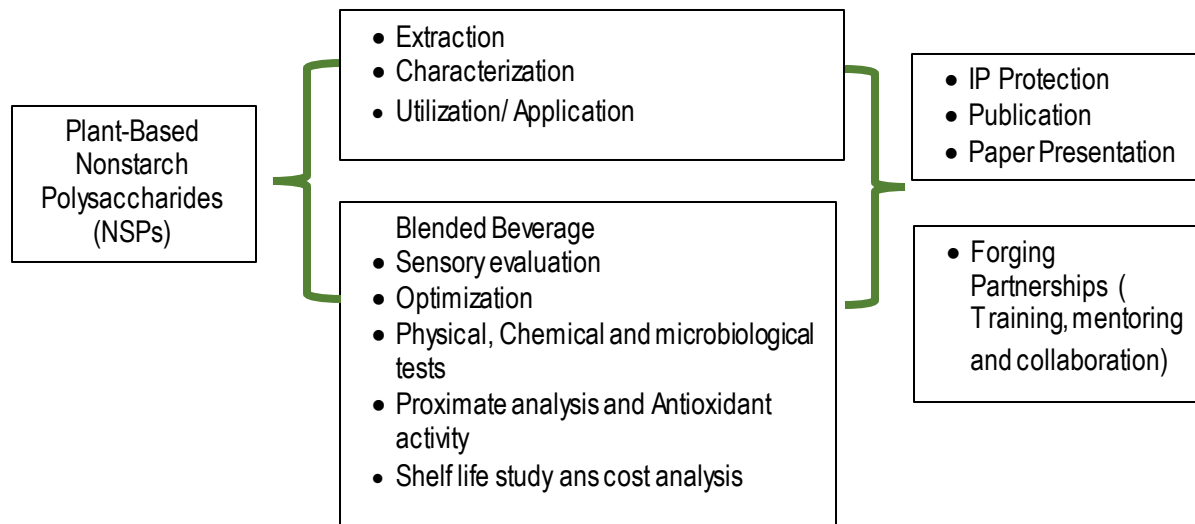
Okra mucilage has been studied as a potential emulsifying agent for food; in addition, its fruit extract is used empirically in traditional cooking to thicken stews and soups. Noorlaila et al. (2015) observed this property in the mucilage extracted from okra fruits of different maturation stages from its inclusion in a coconut milk oil–water emulsion system. The results confirmed the possibility of using okra as an emulsifier in the food industry. Mucilage can improve the quality of food in terms of stability, texture, and appearance, also acting as gelling agents or texture modifiers.

Burbidge et al (2012), invented a Cohesive low-density liquids to stimulate safe swallowing in dysphagic patients wherein the botanical hydrocolloids are selected from the group consisting of plant-extracted gums and plant-derived mucilages. The cohesive liquid used okra gum in combination with kiwi fruit gum. Likewise, Yaqin and Junxiang (2015), disclosed their invention on a kind of women health beverage specially and preparation method thereof, belong to field of health care products. This health beverage is made up of the raw material of following percentage by weight: the water of blueberry juice 30-40%, *Abelmoschus esculentus* juice 30-40%, *Flos Lonicerae* extractive solution 10-15%, *Radix Dauci Sativae* juice 10-20%, citric acid 0.3-0.5%, Mel 3-5% and surplus. However, the manner of juice preparation for okra pods involved crushing of the pods then followed by filtering.

Another invention on functional health care beverage and preparation method and application thereof with Patent No. CN102160668B, was develop by Ke and Yin in 2011. The functional health-care drink is characterized in that being made up of the following material of weight percentage: the tender fruit 20~50% of okra, and the fruit of Chinese wolfberry 5~20%, *Rhizoma dioscoreae* frOm Henan of China 10~20%, honey 5~10%, all the other are pure water.

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(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.)



(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Objective No.1. To determine the NSPs content of okra pod as affected by levels of maturity ;

Sample collection

Okra samples will be procured from the markets in the second district of Capiz. Only those which are commercially mature will be considered as sample in the study.

Mucilage extraction

Okra pods maturity will be 5,7 and 9 days after bloom. Mucilage extraction follows the method of Cahyana et al, (2017) with slight modification. The pods of okra was cut to remove the seeds. Then it will be sliced thinly and soaked in distilled water (1 part okra to 6 parts distilled water) for 12 hours at a temperature of 4-5 °C.The mixture will be heated at 80-86°C for 30 minutes. It will then be allowed to stand undisturbed for 1 hour to to complete the mucilage extraction process. After which, the mixture will be strained using a muslin cloth to separate the okra from the liquid. The liquid extract will be used directly for beverage preparation while for the determination of the functional properties, the liquid mucilage will be dried until contant weight at a temperature of 45°C.The dried mucilage will be weighed to determine which level of okra pod maturity will yield the highest mucilage weight.

Objective No.2.To evaluate the functional properties of NSPs from okra;

Determination of bulk density

The bulk density of the mucilage powder will be determined according to the method described by Gupta et al,(2005). About two grams of mucilage powder will be placed in 10- ml test tube. The test tube will be tapped several times (minimum 10 times) on the laboratory bench to compact the mucilage powder. The final bulk volume will be recorded.

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Determination of water and oil absorption capacity

Water and oil absorption capacities were determined according to the method described by Aremu et al, (2007). One gram of sample will be mixed with 10 ml of distilled water or oil (specific gravity of 0.929) in a centrifuge tube and will be allowed to stand at room temperature for 1 hr. It will then be centrifuged at 200 rpm for 30 min, and the supernatant will be transferred into a 10-ml graduated cylinder.

Determination of emulsifying properties

The emulsifying capacity and emulsion stability of the mucilage of the pods of okra will be determined according to the method described by Thanatcha and Pranee (2011). About 1 g of mucilage powder was dissolved in 50 ml of distilled water, and 50 ml refined oil was added into the mixture. Then, the mixture was homogenized for 1 min and was centrifuged at 2,000 rpm for 5 min. Finally, the height of the emulsified layer was measured and compared with the height of the whole layer. The emulsion stability was estimated after heating the emulsion contained in a calibrated centrifuge tube at 80°C for 3 min in a water bath. The heated emulsion was cooled for 15 min under running tap water and was centrifuged for 15 min.

Determination of foaming properties

Foaming properties including foaming capacity and stability of the mucilage will be determined according to the method of Aremu et al. (2007). One gram of the sample will be dispersed in 50 ml of distilled water. The resulting solution will be vigorously whipped for 30 min in a blender and poured into a 100-ml graduated cylinder. The volume before and after whipping will be recorded, and the foaming capacity will be calculated as percentage volume increase. Foaming stability will be determined as the volume of foam that will remain after 8 hours and will be expressed as a percentage of the initial volume.

Objective No. 3. To optimize the levels of NSPs extracted from okra for blended beverage application;

Experimental Design

The study will follow a factorial experiment and will be composed of 2 factors: Factor A will be the level of maturity (5, 7 and 9 days after bloom) while Factor B is the Levels of mucilage extract (ml). Levels of mucilage extract will be as follows: 100, 200 and 300 ml.

Experimental Treatments

Treatment Combinations for Optimization

Okra Maturity (days after bloom)	Levels of Mucilage Extract (ml)		
	100	200	300
5	1	2	3
7	4	5	6
9	7	8	9

Treatment 1 – 5 days maturity; 100ml mucilage extract

Treatment 2 – 5 days maturity; 200ml mucilage extract

Treatment 3 – 5 days maturity; 300ml mucilage extract

Treatment 4 – 7 days maturity; 100ml mucilage extract

Treatment 5 – 7 days maturity; 200ml mucilage extract

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Treatment 6 – 7 days maturity; 300ml mucilage extract
Treatment 7 – 9 days maturity; 100ml mucilage extract
Treatment 8 – 9 days maturity; 200ml mucilage extract
Treatment 9 – 9 days maturity; 300ml mucilage extract

Beverage Formulation

Formulation of the blended beverage will follow the original formulation of Tanglad-Lupo in Calamansi Juice Except substituting the water with the okra mucilage extract. Processing method will strictly follow the standard procedure.

Objective No.4. To determine the sensory properties and acceptability of NSP-enriched blended beverage;

Sensory Evaluation

The reformulated blended beverage will be subjected to sensory evaluation employing 30 semi-trained panelists. The product acceptability will be evaluated following the 9-point hedonic scale while product characteristics will be evaluated using the quality scoring method. Each product will be assigned a 3-digit code to avoid bias during evaluation. Each panel will be given water to rinse their mouth every after taste.

The evaluation will be carried out mid-morning or mid-afternoon to make sure that the paneslists are neither hungry nor full at the time of tasting.

Sensory attributes to be evaluated include color, aroma, taste, consistency, homogeneity and general acceptability.

Objective No.5 To evaluate the physical, chemical and microbiological characteristics of the formulated beverage

1. Physico- Chemical Composition

The beverages that will be produced from all the experimental treatments will be analyzed for their physico-chemical qualities which include the pH, total soluble solids (TSS), titratable acidity (TA),

pH

The pH of the produced beverage will be measured by a laboratory pH meter. About 20 mL of the sample will be poured in the beaker and pH will be determined by immersing the electrode of the pH meter in it.

TSS

For the total soluble solids (TSS), a drop of the sample will be placed into the lens of the handheld refractometer and read through the eyepiece against the light. The reading will be recorded as °Brix. (Note: refractometer must be set to zero by washing the glass plate with distilled water and blotted dry with a clean and dry cloth).

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Titration Acidity (% Citric Acid)

Two milliliter of the sample will be placed in a 25-ml graduated cylinder and will be diluted to the mark with distilled water, and will be poured to 50-mL Erlenmeyer flask. Three drops of the phenolphthalein as an indicator will be added and titrated with 0.1N NaOH. The volume of NaOH used will be recorded. Titration acidity will be calculated as percent citric acid (predominant acid)

Computation:

$$\% \text{ CA (mg/mL)} = \frac{N_{\text{NaOH}} \times \text{mL}_{\text{NaOH used}} \times \text{meq wt citric acid}}{\text{mL sample}} \times 100$$

where: meq citric acid = 64/1000 = 0.064

2. Color Analysis

The Hunter L^* , a^* and b^* values of powdered oyster mushroom will be measured using a Konika Minolota color reader CR-10 tri-stimulus colorimeter. The dimension L^* means lightness, a^* value indicates redness when positive and greenness when negative and b^* value represents yellowness when positive and blueness when negative.

Microbiological Analyses

Total plate count of samples will be determined using Plate Count Agar (PCA) and Potato Dextrose Agar (PDA). Testing for the presence of *Salmonella* and *E.coli* will also be conducted. This will be done using Brilliant Green Agar, Bismuth Sulfite Agar, Salmonella-Shigella Agar, Xylose-lysine deoxycholate, Eosin Methylene Blue Agar and MacConkey Agar.

Optimum Levels Identification

Contour plots of all the results of the physico-chemical tests, sensory evaluation and color analysis will be superimposed and the region common to all the parameters will be considered as the optimum levels of the experiment. Contour plots will be generated using statistica v.6

Objective No. 6 To determine the antioxidant activity of the optimized blended beverage;

Total Phenolics Content and Radical Scavenging Activity using DPPH assay

To prepare indigenous raw material extract, 10 g of sample will be added with 20mL 80% ethanol and homogenized at high speed for 2 min. The homogenate will be boiled for 5 minutes in water bath and then centrifuged for 10 minutes. The filtrate will be collected and the remaining phenolics in the residue will be extracted using the above method. The filtrate will be pooled and stored at -20°C until analysis of total phenolics and antioxidant activity.

Total phenolic content will be determined using the Folin-Ciocalteu method following the procedure of (Lagnika et al, 2014). Extract will be diluted by adding 0.9 mL distilled water to 0.1 mL extract. Then 2 mL of 0.25 N Folin-Ciocalteu phenol reagent will be added. The solution will be mixed and allowed to stand for 5 minutes at room temperature. Two ml of 1N Na_2CO_3 solution will be added and mixed. The solution will be allowed to stand for 1 hr at room temperature. The absorbance of the final solution will be recorded with spectrophotometer at 640 nm with respect to blank solution (80% ethanol). The result will be expressed as mg of gallic acid equivalent per g of fresh weight (mg GAE.g⁻¹).

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5. To evaluate the physical, chemical and microbiological characteristics of the formulated beverage.												
6. To determine the antioxidant activity of the optimized blended beverage; and												
7. To assess consumer acceptability of the optimized product.												
8. To conduct shelf life studies of the Optimized product												
9. Manuscript preparation, writing of terminal report and drafting of patent application												
10. Forging partnerships with potential collaborators												

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

- 1 optimized NSP-enriched blended beverage
- 1 IP protected product
- 1 journal article
- 1 IEC material
- 1 forge partnership with potential technology adapter

(16) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

(17) Target Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

Beverage and food processors in the province of Capiz, Students, Farmers, Entrepreneurs and Consuming public

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(18) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
MARIFE R.HILAPAD	Study Leader	Food Science and Technology
	Research Assistant	Food Technology

Percent time devoted to the Program/ Project/ Study: 20%

Program/Project Leader (in hours): 8hrs/wk

Program/Project Members (in hours): 40 hrs/wk

(19) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)	1 @11,068/mo for 12 months	132,816.00
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Transportation allowance	10,000.00	10,000.00
- Representation (sensory evaluation and consumer testing)		5,000.00
- Raw Materials	3@1,200/tank	50,000.00
- LPG Refill		3,600.00
- Thermometer		5,000.00
- Professional fees (laboratory analysis and other related fees) DPPH Assay	20,000.00	20,000.00
Reagents		
Acetone	3@ 2500/2.5L	7,500.00
Follen-ciocalteu	12,000/bot	12,000.00
Petroleum ether	3,500/bot	3,500.00
Culture media		20,000.00
Miscellaneous (for incidental expenses)		11,034.00
PET Bottles	500@3.50/pc	1,750.00
Muslin cloth		1,500.00
Heat gun		7,000.00
Patent Application fee		10,000.00
Journal fee		25,000.00
C. CAPITAL OUTLAY		
Water bath		75,000.00
Double Boiler		60,000.00
Chiller		60,000.00
Total		520,700.00

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(21) Capsule Curriculum Vitae (One-page brief C.V. only)

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Designations

- Program Chair, Bachelor of Science in Food Technology
- Project Leader, DOST- PCIEERD Funded Project
(QUALISAFE LAB)
- Project Staff, Regional Agri- Aqua Technology Business Incubator

Educational Background

Level	Name of School	Degree / Course	Year Graduated	Inclusive Dates of Attendance	Scholarship
Doctorate	University of the Philippines Los Banos (UPLB)	PhD Food Science	2018	2015 - 2018	DOST-ASTHRDP
Masters	Visayas State University	MS Food Science and Technology	2013	2011-2013	DOST-ASTHRDP
College	Visayas State College of Agriculture (now VSU)	BS in Food Technology	1994	1990-1994	-

Eligibility

Title of Exam / Competency	Rating	Date of Exam	Place of Exam
Civil Service Exam (Professional Level)	87.03 (Ranked 2)	November 7, 1999	Sogod, Southern Leyte, Philippines
National Certificate II in Commercial Cooking		July 9, 2009	TESDA Capiz
National Certificate II in Food and Beverage Services		September 21, 2007	TESDA Capiz

Work Experience

Inclusive Date	Designation	Agency/Office
February 3, 2003 to present	Faculty	Capiz State University

Publications

Title of Research	Journal	Date of Publication
1. Optimization of processing parameters for vacuum fried oyster mushroom (<i>Pleurotus ostreatus</i> (Jacquin) P. kummer)	Food Research 4 (4): 1371 – 1382 DOI: https://doi.org/10.26656/fr.2017.4(4).065	April 28, 2020